

Super Store Sales Analysis: An Interactive Business Intelligence Dashboard Using Microsoft Power BI

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Abstract—This study analyzes retail Super Store sales data using Microsoft Power BI. It follows a data analytics pipeline including preprocessing, analysis, and visualization. Key insights show Technology as the top category, Consumer segment contributing 51% of revenue, California and New York as leading states, and peak sales in Q4 2020. An interactive dashboard with charts and maps enables real-time business insights and decision-making.

Index Terms— Power BI, Sales Analytics, Data Visualization, Dashboard

I. INTRODUCTION

A huge volume of transactional data is produced daily within the retail industry. The ability of businesses to capitalize on this information is central to optimizing business operations, profitability, and customer experience. Business Intelligence tools such as Microsoft Power BI have reshaped the business analytical environment by offering a means for both presenting and interacting with data to allow for objective-based, data-driven decision making in the form of interactive dashboards.

In this paper, we analyze a Super Store sales dataset from 2019-2020 containing the transaction details, customers and products (order and category), shipments and geographical information, payments and payment types, sales amounts, and profit. Through the use of structured analytics, this paper seeks to extract meaningful business patterns and insights which lead to objective based business decisions.

The aims of this paper are to (1) thoroughly clean the dataset, (2) complete five unique objective-based analyses, (3) build a completely interactive dashboard using Power BI and (4) create business actionable insights. Sections II through VII describe the dataset, preprocessing steps, analyses, dashboard, and results respectively.

II. DATASET DESCRIPTION

Super Store is a famous retail dataset that includes 9800 transaction records across the US for fiscal years 2019 & 2020. It has 21 variables representing order, customer,

product and geographical features along with performance indicators (financial results). Table I shows the most significant features taken in consideration for this analysis.

TABLE I. KEY FEATURES OF THE SUPER STORE DATASET

Feature	Type	Description
Order ID	String	Unique order identifier
Order Date	Date	Order placement date
Ship Date	Date	Shipment dispatch date
Ship Mode	Categorical	Shipping method used
Segment	Categorical	Consumer/Corporate/Home Office
State	String	U.S. state of customer
Category	Categorical	Product main category
Sub-Category	String	Product sub-category
Sales	Numeric	Revenue from sale (USD)
Profit	Numeric	Profit earned (USD)
Payment Mode	Categorical	COD / Online / Cards

Table I: Dataset feature overview with data types and descriptions.

The dataset includes the four regions (Central, East, South, West) and all states within the U.S. Products belong to 3 broad categories (Furniture, Office Supplies, Technology) that contains 17 sub-categories. Performance are mainly evaluated on Sales and Profit in all the analysis.

III. DATA PREPROCESSING

All data cleaning and preparation was done in Power BI using the Power Query Editor, to ensure the accuracy and validity of analytical findings before they are visualized.

A. Missing Value Treatment

All columns were scanned for null values in the dataset. The rows with the null values for the Order Date and Ship Date were removed because the Order Date and Ship Date are important for temporal analysis. Postal Code values were filled using the mode of the states.

B. Data Type Conversion

The data columns for date (Order Date, Ship Date) were also changed to the correct data type (date) within Power Query so that Ship Days (difference between Ship

date and Order date) can be calculated and a Month, Year to show a trend.

C. Feature Engineering

Generated calculated features: Created 'Year' column from 'Order Date' to allow for Year over Year comparisons, 'Month' column for monthly trend chart and 'Ship Days' which is the difference between 'Ship Date' and 'Order Date' (an integer value) to allow for examination of delivery speed.

D. Duplicate Removal and Normalization

The Order ID field was checked and all duplicate entries removed to avoid overcounting. The labels in the categorical features (Ship Mode, Segment, Payment Mode) were made uniform and the whitespace from all text fields was stripped.

IV. OBJECTIVE-BASED ANALYSES

Five structured objective-based analyses were conducted to extract actionable insights. Each analysis addresses a specific business question and is visualized within the interactive dashboard.

A. Analysis 1: Sales by Product Category and Sub-Category

The objective of this analysis is to identify which product categories and sub categories can generate the most revenue to allow for the management of inventory, thus buying decisions can be made based on priority. Technology is found to be the highest category yielding 0.12M followed by office supplies which generated 0.11M and furniture at 0.10M. At the sub-category level, phones yielded 0.12M, binders yielded 0.11M, and chairs yielded 0.10M. Based on this result, it is highly profitable to market the products in the technology category.

B. Analysis 2: Regional and State-Level Sales Distribution

Objective: To identify geographic sales patterns and highlight high-value markets for regional resource allocation and expansion planning.

California dominated state-level sales with 19%, followed by New York at 17%, Pennsylvania and Texas at 6% each, and Florida and Ohio at 4% each. The concentration in coastal states indicates that Eastern and Western markets are primary revenue sources, while Central and Southern states represent growth opportunities.

C. Analysis 3: Year-over-Year Monthly Sales and Profit Trends

Objective: To compare monthly sales and profit performance across 2019 and 2020 to identify seasonal patterns, growth trends, and anomalies.

Both years exhibit consistent growth from January through December, with a pronounced spike in November–December driven by holiday demand. Sales in 2020 were notably higher across most months compared to 2019, indicating year-over-year growth. Profit trends mirrored sales with Q4 displaying the highest margins, confirming that proactive Q4 inventory stocking is strategically critical.

D. Analysis 4: Customer Segment Profiling

Objective: To understand revenue distribution across Consumer, Corporate, and Home Office segments to guide targeted marketing strategies.

The Consumer segment holds the largest share at 51%, followed by Corporate at 33%, and Home Office at 17%. Despite being the smallest, the Home Office segment represents an emerging market in the post-2020 remote-work era, suggesting product bundling opportunities could unlock additional revenue.

E. Analysis 5: Payment Mode and Shipping Behavior

Goals: Examining customer payment and shipping habits in order to improve payment gateway and logistic management.

Most payments are Cash on Delivery with 42%, followed by Online 34% and Cards with 24%. In regard to shipping, Standard Class wins by a margin with 0.33M, followed by Second Class with 0.11M, First Class 0.08M and Same day 0.03M. The average duration to ship is 7 days. With such a high rate on COD it implies there are trust barriers for online payments and the company could solve it using loyalty rewards system.

V. INTERACTIVE POWER BI DASHBOARD

A interactive dashboard called "Super Store Sales Dashboard" was created in Microsoft Power BI Desktop. The dashboard's theme is set to dark theme to help readers see the graphics clearly and bring all findings of analysis into one view to the stakeholder in the image of Fig. 1.



Fig. 1. Super Store Sales Dashboard in Microsoft Power BI Desktop.

A. Dashboard Components

Components of the dashboard

Total of eleven components on the dashboard are (1) KPI Card (Sales, Profit, Orders, Ship Days) which can be dynamically updated by selection of filter(s) on dashboard; (2) regional Page Level Slicer Button (select Central, East, South, or West region) (3) sales by states pie chart (4) Sales By Month YoY Dual year area line chart; (5) Sales by ship mode horizontal bar chart (6) Sales by segment donut chart (7) Sales by Month profit YoY area line chart (8) sales by category bar chart (9) sales by payment mode donut chart (10) sales by sub category bar chart (11) Bing map geographically visualizing sales at state level with colors:

B. Cross-Filtering and Interactivity

All visuals are cross-filtered, meaning that selecting any data point in one visual automatically updates all others to reflect the corresponding data subset. This enables drill-down analysis without requiring technical knowledge, making the dashboard accessible to non-technical business stakeholders.

VI. RESULTS AND DISCUSSION

The analyses yield several significant business insights. Table II summarizes the key findings across all five objective-based analyses.

TABLE II. SUMMARY OF OBJECTIVE-BASED ANALYSIS RESULTS

Analysis	Key Finding	Value
Sales by Category	Technology leads	0.12M
Top Sub-Category	Phones dominate	0.12M
Ship Mode	Standard Class tops	0.33M
Top State	California leads	19%
Customer Segment	Consumer majority	51%
Payment Mode	COD most used	42%
YoY Growth	2020 > 2019	Q4 surge
Avg Ship Days	Overall average	7 days
Profit (filtered)	Sample profit	1.91K

Table II: Summary of key findings across all five analyses.

Phones, the category that falls under "technology products" is the largest money-making segment showing that good inventory management and pricing are key. California and New York dominance is the result of a robust distribution channel for both areas.

The YoYo trend shows that 2019 and 2020 follow the same pattern of growth and Q4 is by far the most important selling period of the year as the largest sales peaks were seen on both year. Majority of 51% for "consumer" segment shows that the marketing strategy should focus on B2C. 42% of COD indicates customer hesitancy to adopt digital payment channels. Cash back/ loyalty program should be implemented to overcome that hesitancy.

The average ship duration of 7 days can be considered as a source for competitive differentiation. By reducing the shipping time with similar pricing, preference can be shifted towards better profit yielding shipping modes i.e. "First Class" and "Same Day".

VII. CONCLUSION

In this document a business intelligence project based on the sales data of Super Store using Microsoft Power BI is analyzed. It was structured in five analyses which were based on objectives of; Product Performance, geographical sales distribution, temporal distribution, Customer Segmentation, payment & shipping behavior. With these, an interactive business intelligence interface suitable for stakeholders, which combines the information is provided. The observations are; Technology has the highest product sale; Consumer segment gives 51% of the revenue; California & New York are the top grossing markets; Q4 is the best grossing quarter; the major payment method is

Cash On Delivery. Further scope is of implementing predictive machine learning models.

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